

Project Initialization and Planning Phase

Date	15 september 2026
Team ID	xxxxxxx
Project Title	<i>India agriculture crop production analysis</i>
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	The primary objective of this project is to analyze India's agricultural crop production data and develop an interactive Tableau dashboard to identify production patterns across different states, crops, years, and seasons. The project aims to transform raw agricultural data into meaningful visual insights that support easy comparison and analysis.
Scope	The project focuses on analyzing historical agricultural crop production data from India. It covers state-wise, crop-wise, year-wise, and season-wise production analysis using Tableau. The project includes data preparation, visualization, dashboard development, interactive filters, and a Tableau Story. The scope is limited to analysis and visualization of the available dataset and does not include real-time agricultural monitoring or crop disease prediction.
Problem Statement	
Description	Agricultural production data contains information about different states, crops, years, and seasons, but analyzing large amounts of tabular data manually can be difficult and time-consuming. It can be challenging to identify high-producing states, major crops, yearly production trends, and seasonal production patterns from raw data. Therefore, there is a need for an interactive visualization system that presents agricultural production information in a simple and understandable manner.
Impact	An interactive visualization of agricultural production data can make

	complex information easier to understand and compare. It can help users identify production trends, compare states and crops, observe seasonal patterns, and obtain insights from the available data. This can support data-driven analysis and improve the understanding of India's agricultural production patterns.
Approach	The proposed solution uses Tableau to analyze and visualize an agricultural crop production dataset obtained from Kaggle. The dataset is connected to Tableau and prepared for analysis by checking the fields and organizing the required dimensions and measures. Various visualizations are created, including state-wise production, crop-wise production, year-wise production trends, season-wise production, and a state production map. Interactive filters for State, Year, Crop, and Season are provided. These visualizations are combined into an interactive dashboard and Tableau Story to present the findings clearly.
Key Features	• Interactive agricultural production dashboard • State-wise production comparison • Crop-wise production analysis • Year-wise production trend analysis • Season-wise production analysis • Geographical visualization using a state map • KPI cards for important production metrics • Interactive filters for State, Year, Crop, and Season • Tableau Story containing multiple analytical views • Interactive chart filtering for exploring relationships between visualizations

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Laptop with standard multi-core processor suitable for Tableau Desktop and data visualization.	Intel Core i5 13th Gen processor; dedicated NVIDIA GPU available; used for Tableau Desktop visualization workloads.
Memory	Minimum 8 GB RAM recommended for Tableau Desktop and dataset analysis	16 GB RAM.

Storage	Sufficient local storage for the Kaggle dataset, Tableau workbook, exported dashboard images/PDF, and project documentation. and logs	1 TB SSD; sufficient for the approximately 28 MB dataset, Tableau workbook, extracts, and project files.
Software		
Frameworks	Tableau Desktop 2026.2	Tableau Desktop and Tableau Public.
Libraries	Not applicable – the project primarily uses Tableau Desktop for data preparation, analysis, and visualization.	Not required; analysis and visualization are performed primarily in Tableau.
Development Environment	Tableau Desktop on Windows, with Google Drive and GitHub used for project file storage and submission.	Tableau Desktop, Tableau Public, Microsoft Excel, and web browser.
Data		
Data	Source: Kaggle agricultural crop production dataset. Format: Excel (.xlsx). Data Used: The dataset contains agricultural production information used for analysis by State, Crop, Year, and Season . Purpose: The data was used to analyze agricultural production patterns and create interactive visualizations and dashboards in Tableau.	e.g., Kaggle dataset, 10,000 images